

ANSWERS

1.

Question	Answer	Marks
6(a)	(frequency =) the <u>number</u> of wavelengths	B1
	(passing a point) per <u>unit</u> time	B1
6(b)	yes and within audible range (of human with healthy hearing)	B1
6(c)	it / the loudspeaker / cone / coil is vibrating	B1
	when it / cone moves (forwards) it produces a compression / compresses (air) or when it moves (backwards) it produces a rarefaction / expands (air)	B1
	vibrations / compressions / rarefactions / energy / <u>change</u> in pressure / motion passed on (to neighbouring air)	B1
6(d)	space is a vacuum or no medium / air in space	C1
	sound cannot travel in a vacuum / space / without a medium	A1

2.

Question	Answer	Marks
4(a)	equipment e.g. (evacuable) container and (suspended electric) bell / source of sound and (air) <u>pump</u>	B1
	action e.g. ringing / sound (produced) / sound heard and evacuate air / vacuum (produced)	B1
	observation e.g. transmission / hearing of sound ceases / no sound	B1
4(b)	(they) vibrate	M1
	(vibration) parallel to / in the direction of propagation / of energy travel	A1
4(c)	(point where) molecules are further apart (than average) or (point where) pressure / density is lower (than average)	B1
4(d)	$(\lambda =) v / f$ or 1500 / 140 000	C1
	1.1×10^{-2} m or 1.1 cm or 11 mm	A1

3.

Question	Answer	Marks
9(a)(i)	(sound that) cannot be heard	C1
	(sound with) frequency above 20 kHz	A1
9(a)(ii)	(some) reflect	B1
	(some) pass into new material or (some) absorbed by material	B1
9(a)(iii)	$(d =) s \times t$ in any form numerical or algebraic	C1
	$1500 \times 0.03 (\times 10^{-3} / 2)$	C1
	0.022 m or 0.023 m	A1
9(a)(iv)	gases 150 – 1000 m/s	B1
	solids 2000 – 8000 m/s	B1

9(b)(i)	(more) X-rays make the image black(er)	B1
	X-rays absorbed by bone or do not reach detector through bone	B1
	<u>some</u> X-rays pass through tissue or tissue absorbs <u>some</u> X-rays	B1
9(b)(ii)	$(f =) v / \lambda$ in any form numerical or algebraic	C1
	1.5×10^{17} Hz	A1
9(b)(iii)	(X-rays) cause mutation / cancer / kill cells / ionisation	B1

4.

Question	Answer	Marks
5(a)	vibration / oscillation of <u>particles</u> or energy transfer or compressions and rarefactions	B1
	vibration / oscillation (of particles) parallel to energy travel direction / propagation direction	B1
5(b)(i)	$(\lambda =) v / f$ or 1500 / 42 000	C1
	0.036 m or 3.6 cm	A1
5(b)(ii)	(ultrasound) vibrates (the cleaning fluid / jewellery)	B1
	dirt shaken off	B1

5.

10(b)(i)	tape measure or trundle wheel or microphones	B1
	stopwatch or timer	B1
10(b)(ii)	start timer on seeing the smoke from gun / sound picked up by microphone	B1
	stop timer on hearing sound	B1
	measure (large) distance between students	B1
10(b)(iii)	both between and including 1000 and 10 000 m/s	B1
	solid speed > liquid speed	B1

6.

Question	Answer	Marks
9(a)	frequency of sound wave small(er) or its frequency is less than 20 000 Hz	B1
9(b)(i)	transmission of energy	B1
	(through a medium) with no net movement of medium or by vibrating particles	B1
	vibrations parallel (and antiparallel) to wave / energy travel direction or cannot be polarised	B1
9(b)(ii)	1 two centres of rarefactions labelled R	B1
	2 distance from one point to adjacent identical point indicated (with double-headed arrow)	B1
	3. $(v =) f\lambda$ or $25\,000 \times 0.047 / 0.048 / 0.049$ or $25\,000 \times 4.7 / 4.8 / 4.9$ or $25\,000 \times 47 / 48 / 49$	C1
	1200 m / s	A1
9(c)(i)	decreases	B1
9(c)(ii)	four / five straight lines in air that touch the compressions still in the liquid and no intermediate / extra lines between the correct lines	B1
	at least four compressions in air parallel to each other	B1
	at least four straight lines at shallower angle from horizontal and slope correct	B1

9(d)	object (to be cleaned) immersed in liquid / solvent	B1
	object / liquid agitated / vibrated by ultrasound	B1
	dirt (particles) shaken off or dislodges / removes dirt	B1

7.

Question	Answer	Marks
6(a)	reflection of <u>sound</u>	B1
6(b)(i)	decreases	B1
6(b)(ii)	does not change	B1
6(c)	$(\lambda =) c/f$ or 330 / 3700	C1
	0.089 m or 8.9 cm or 89 mm	A1

8.

Question	Answer	Marks
5(a)	pressure / vibrational / longitudinal (wave) or (wave that consists of) compressions and rarefactions or sound <u>wave</u> or inaudible sound	B1
	<u>frequency</u> greater than 15 kHz–25 kHz	B1
5(b)	(ultrasound) transmitted into body	B1
	echo / reflection from (baby / fetus)	B1
	image produced	B1

9.

Question	Answer	Marks
6(a)	longitudinal - vibration / oscillation / movement to and fro and in direction of wave or has compressions and rarefactions	B1
	transverse – vibration / oscillation / movement up and down and at right angles to wave or has crests and troughs	B1
6(b)(i)	$(\lambda =) v / f$ or 330 / 3800	B1
	0.087 m or 8.7 cm	B1
6(b)(ii)	not heard and because below the range of audible frequencies or audible range is 20 – 20 000 Hz or too low a pitch / frequency	B1

- 10.
- (a) number of oscillations / vibrations / wavelengths / compressions / rarefactions / cycles per second / unit time B1
- (b) (i) $(\lambda =)c/f$ or $330/2200$ C1
0.15 m A1
- (ii) 1 no change
and
2 increases B1
- (c) (i) 1 loudspeaker vibrates / oscillates / moves to and fro (and collides with molecules) B1
2 compressions **and** rarefactions / molecules vibrate / longitudinal wave vibration / oscillation / energy passed on B1
B1
- (ii) fewer / no molecules / particles **and** less / no energy / vibration transferred B1
- 11.
- (a) at compression: molecules closer together **or** pressure higher **or** vice versa for rarefaction B1
- (b) (i) $v = f\lambda$ **or** in words B1
- (ii) larger **and** because the frequency is lower B1
- (c) states one use (e.g. prenatal scanning) B1
basic idea (e.g. ultrasound reflects off foetus) B1
- 12.
- (a) longitudinal / pressure / sound (wave) **or** compressions **and** rarefactions B1
(frequency) greater than 15 – 25 kHz / above limits of audibility B1
- (b) $(x =)vt/2$ **or** $340 \times 0.030/2$ **or** 340×0.015 **or** 10.2 C1
5.1 m A1

- 13.**
- (a)** (shortest) distance between two points in phase [B1]
- (b) (i)** oscillate **or** vibrate **or** move closer and further [B1]
backwards and forwards **or** in direction of wave (energy) **or** longitudinal [B1]
mentioned
- (ii)** move in opposite directions **or** when A is on right B is on left (of mean) [B1]
or A molecules next move apart **and** B next move together
- (c)** diagram containing sound source (bell/tuning fork, etc.) in container [B1]
vacuum/pump connected to container [B1]
remove air [M1]
sound decreases [A1]
- (d) (i)** sound (that returns) after a reflection [B1]
- (ii)** ($s =$) d/t in any algebraic or numerical form e.g. 20 (or 40)/ 0.12 (or 0.06) [C1]
 330 **or** 333 m/s [A1]
- (iii)** ($f =$) v/λ in any algebraic or numerical form e.g. $330/30$ (or 0.03) [C1]
correct conversion of 30 mm to 0.03 m [C1]
 $11\,100$ Hz **or** $11\,000$ Hz [A1]
- (iv)** 0.015 m **or** 15 mm [B1]
- 14.**
- (a)** transmission of energy through a medium **or** vibration **or** oscillation **or** [C1]
two opposite motions (e.g. up and down) **or** compressions and rarefactions [A1]
vibration direction parallel to energy travel/wave direction **or** similar
- (b) (i)** $1.5\text{--}2.5 \times 10^4$ Hz **or** $15\text{--}25$ kHz **cao** [B1]
 $15\text{--}25$ Hz **cao** [B1]
- (ii)** ($\lambda =$) c/f **or** 330 /either of candidate's frequencies [C1]
 330 /candidate's higher frequency and correctly calculated with unit
(candidate's higher frequency is either the one stated as the highest
or the one that is in fact the higher) [A1]

15.		
(a)	reflection (of sound/ultrasound)	B1
(b)	waves of same period (by eye)	B1
	smaller amplitude (by eye)	B1
(c)	(i) 20–20 000 Hz	B1
	(ii) higher than (i)	B1
16.		
(a)	how sound is made e.g. gun, clap hands, hit metal	B1
	correct measurement of time, e.g. from seeing flash to hearing sound, clap-echo	B1
	correct measurement of distance, e.g. gun to observer, observer to wall	B1
	correct calculation for measurements, e.g. d/t or $2d/t$	B1
	precaution e.g. time clap on echo and time 10; ensure no wind; repeat in opposite direction; repeat and average; use large distance; use more than 200m	B1
(b)	(i) (sound/wave/vibration) of high frequency or (sound that) cannot be heard	C1
	(frequency) above 15–20 kHz	A1
	(ii) $f = 1/T$ or 6×10^{-6} (s) seen or 2,3,4 pulses in 12,18,24 μs	C1
	1.7×10^5 allow 166667	A1
(iii)	not all sound is reflected (from back surface) or some passes through the back (some energy/sound) absorbed (by metal) (sound/energy) spreads out/scattered/reflected in other directions/dispersed/ travels a (greater) distance	B2
(iv)	at least one pulse half way between S and R in the long gap	B1
	total height of pulse smaller than S and 3 or more drawn and labelled C	B1
(v)	$v = f\lambda$ in any algebraic or numerical form e.g. $4000/8 \times 10^6$	C1
	$5(.0) \times 10^{-4} \text{ m}$	A1

17.		
(a) (i)	C in correct position i.e. gap 4, 18 or 32 { allow arrows/ R in correct position i.e. gap 11 or 25 { brackets < $\lambda/2$ OR two correct positions but R and C reversed 1/2	B1 B1
(ii)	6.2 $\rightarrow$ 6.6 cm	B1
(iii)	($v =$) $f\lambda$ or $5.1/5100 \times 6.4/0.064$ (using candidate's 5 (a) (ii)) $3.16 - 3.37 \times 10^n$ 316 – 337 m/s	C1 C1 A1
(b)	(longitudinal wave:) vibration/oscillation direction parallel to/in same direction as wave/energy travel direction (not L & R) transverse wave: directions perpendicular or can be polarized (not up & down)	B1 B1
18.		
(a)	(sound) too high a frequency to be heard or (frequency) above 20 kHz	B1
(b)	($f =$) v/λ or $v = f \lambda$ algebraic or numerical 1 250 000 Hz	C1 A1
(c)	vibrate/oscillate vibration etc. in same direction as/parallel to wave/energy or horizontally	C1 A1
(d)	pressure increases and decreases or compressions and rarefactions mentioned in (d) or particles come together and move apart	B1
19.		
(a)	cone/molecules vibrate molecules (vibrate) longitudinally/back and forward (in direction of sound) or compressions and rarefactions mentioned (e.g. longitudinal waves)	B1 B1
(b) (i)	a number from 18,000 to 22,000 Hz	B1
(ii)	($v =$) $f \lambda$ algebraic or numerical using 20 Hz or candidate's (i) 17 m	C1 A1
20.		
(a)	distance traveled per unit time or in one second / distance $\div$ time or rate of change of distance	B1
(b)	$s = d/t$ in any algebraic or numerical form any doubling of distance or final time 0.48 s	C1 C1 A1
(c)	60/0.48 (5) 123.75 accept 120, 123, 124	C1 A1
	(allow 0.24s 2/3 accept 0.5s) (ecf (b))	